

Representation Type and Readout Expression in Neural Group Composition

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Abstract

We train one-hidden-layer MLPs to complete the multiplication table of a finite group, holding group order, dataset size and output cardinality fixed so that only the algebra varies. We report two findings.

(A) Groups carrying quaternionic-type irreducible representations reach substantially lower validation accuracy at a fixed data budget and a fixed training budget than groups that do not, at fixed order. We test this on the complete population of all fifteen groups of order 24, which removes any discretion over which groups to include. Ranked by mean final validation accuracy over twelve independent instances each, the five quaternionic-type groups are separated from the other ten by a preregistered exact test at $p = 1/\binom{15}{5} = 3.3 \times 10^{-4}$. The sharpest comparison is the character-table twin: pairs such as $D_8 \times \mathbb{Z}_3$ and $Q_8 \times \mathbb{Z}_3$ that share their entire ordinary character table but differ in Frobenius–Schur indicator, hence in the real structure of their group algebras ($M_2(\mathbb{R})$ versus $\mathbb{H}$). There are two such pairs at order 24 and both separate in the predicted direction. A third pair matched only on irreducible degrees and class sizes, preregistered before it was trained, separates too.

(B) A frozen hidden layer in $Q_8 \times \mathbb{Z}_3$ supports a linear readout far better than the one the network trained; but how much better is not a property of the network. Fitting thirty readout configurations to the same frozen representation — the same representation, within each seed — the reported gap runs from -0.021 under a recently published probe protocol to $+0.235$ under ridge with the regulariser chosen inside the training split, a range of 0.256 . Holding objective, optimiser, budget and initialisation fixed and varying only the readout weight decay still moves it from $+0.002$ to $+0.181$. Under the published protocol the character-table twins do not separate; under ridge they do ($+0.235$ against -0.033). We argue the linear-probe gap is not well defined without stating the probe’s regularisation.

A factorial over end-to-end and readout weight decay separates two roles that have been conflated: higher end-to-end weight decay is associated with far greater linear decodability at a fixed budget (probe accuracy $0.162 \rightarrow 0.992$), while readout weight decay limits how much of it a linear readout expresses (span 0.146 at fixed representation).

We state explicitly what is not established, including any causal chain from Frobenius–Schur type to the readout defect, and the fact that character-table matching does not isolate the Frobenius–Schur indicator.

1 Introduction

Grokking — delayed generalisation long after training loss saturates — has been characterised largely in terms of optimisation quantities: weight decay, learning rate, norm ratios, circuit efficiency [1, 2, 5]. Recent work sharpens this. Truong [10] shows that under cross-entropy the weight-norm dependence of the delay runs through the effective logit scale rather than the scalar norm, and Chou

et al. [6] decompose networks into encoder and linear readout and give four diagnostic signatures for grokking and double descent. The circuit literature on group composition asks which algorithm forms [3, 4], not when it forms or why some tasks resist it. Notsawo et al. [7] widen the task family to finite-dimensional algebras and show that associativity, unitality, commutativity and structure-tensor statistics affect grokking delay and sample complexity.

What is largely absent is a controlled variation of the algebraic structure of the task at fixed cardinality. This is precisely what the finite-dimensional algebra framework cannot reach within groups of a fixed order: there the structure tensor is $\mathcal{C}_{ijk} = \mathbb{1}(g_i g_j = g_k)$, whose mode-3 rows have pairwise disjoint supports of size n , so $\text{rank } \mathcal{C}_{(3)} = n$ and the sparsity is n^2 for *every* group of that order. Associativity and unitality hold by the group axioms, and commutativity agrees within every twin pair. Unfolding rank and canonical-basis sparsity are therefore constant across every group of a fixed order, and within the genuine twin pairs all of the coarse properties they study coincide — though commutativity does vary across the fifteen groups. The outcome varies by more than 0.9 of accuracy.

This work varies the group at fixed order, dataset size and output cardinality. The central instrument is the character-table twin: two groups whose ordinary complex character tables are identical, so that essentially every coarse invariant a network could plausibly be sensitive to — number of conjugacy classes, irreducible dimensions, commuting probability — is matched by construction. What differs is the real structure — though not *only* that: the twins also differ in their power maps, involution counts, subgroup lattices and element-order profiles, all of which a network can read off the Cayley table (§8). For D_8 and Q_8 ,

$$\mathbb{R}[D_8] \cong \mathbb{R}^4 \oplus M_2(\mathbb{R}), \quad \mathbb{R}[Q_8] \cong \mathbb{R}^4 \oplus \mathbb{H}. \quad (1)$$

Both $M_2(\mathbb{R})$ and $\mathbb{H}$ complexify to $M_2(\mathbb{C})$, which is exactly why the ordinary character table cannot distinguish them. The Frobenius–Schur indicator can, because it requires $\chi(g^2)$ — the power map — which the character table as a matrix does not contain. Equivalently, the two differing simple blocks $M_2(\mathbb{R})$ and $\mathbb{H}$ — not the full semisimple algebras — represent the two classes of $\text{Br}(\mathbb{R}) \cong \mathbb{Z}/2$.

Contributions.

1. A matched-character-table design evaluated on the *complete* population of order-24 groups, which removes within-order group-selection discretion (§2, §3).
2. Evidence that representation type predicts validation accuracy at a fixed data and training budget, stated at the correct unit of replication (§3).
3. A demonstration that the linear-probe gap depends on the probe, spanning 0.256 across recipes on a fixed representation and 0.179 when only the readout weight decay is varied, with a published protocol landing on the side reporting no gap (§4).
4. A factorial separating two opposite roles of weight decay (§5).
5. An exact isotypic decomposition of the output map, with transplantation under a donor fixed in advance (§6).
6. A threats-to-validity section naming what the design does not isolate (§8).

2 Setup

Task. For a finite group G with $|G| = n$, the model receives $(g, h) \in G^2$ as the concatenation of two one-hot vectors in $\mathbb{R}^{2n}$ and predicts gh as one of n classes. All groups have $n = 24$, so the dataset is 576 pairs and the output space is 24 classes in every condition. A fraction ρ of pairs is used for training and the remainder for validation.

Model and optimisation. One hidden layer, width 512, ReLU, no biases; full-batch AdamW, $\text{lr} = 2 \times 10^{-3}$, $\beta = (0.9, 0.98)$, weight decay 1.0 unless swept, cross-entropy.

The population. There are exactly fifteen groups of order 24 up to isomorphism. The Small Groups Library lists fifteen isomorphism classes of order 24; we construct fifteen pairwise non-isomorphic groups matching that count, separated by a fingerprint combining element-order profile, conjugacy-class sizes, isotypic sector types, centre order and derived-subgroup order. Distinct fingerprints establish at least fifteen; the upper bound is the classification result. Using the whole population removes within-order group-selection discretion — though not the choice of order 24, of ρ , of budget, or of final accuracy as the outcome. Five carry quaternionic-type irreducible representations — $\text{SL}(2, 3)$, Dic_{24} , $Q_8 \times \mathbb{Z}_3$, $\mathbb{Z}_2 \times \text{Dic}_{12}$ and $\mathbb{Z}_3 \rtimes \mathbb{Z}_8$ — and there are exactly two character-table twin pairs.

Cayley tables are checked programmatically: unique identity, existence of inverses, exhaustive associativity over all $24^3 = 13,824$ triples, and the Frobenius–Schur identity $\sum_{\rho} \varepsilon_{\rho} d_{\rho} = \#\{g : g^2 = e\}$.

Instances. Each of the twelve instances per group draws an independent random relabelling of the group, an independent train/validation split and an independent initialisation. This separates three sources of variance that a single seed would confound. With one-hot inputs, no biases and i.i.d. initialisation, relabelling is a permutation of the coordinate axes and is distributionally equivalent to resampling the split; we run it anyway as a check on that argument.

Isotypic projectors. At $\rho = 0.60$ the split is 345 training and 231 validation pairs; at $\rho = 0.70$ it is 403 and 173. Class sums span the centre of $\mathbb{R}[G]$. A random symmetric central element is scalar on each real isotypic component, so its eigenspaces are those components; conjugate pairs of complex-type irreducible representations merge, which is what a real network sees. Each The Frobenius–Schur indicator of an irreducible character is $\nu_{\chi} = |G|^{-1} \sum_g \chi(g^2) \in \{+1, 0, -1\}$. The quantity we attach to a real isotypic sector is $s = |G|^{-1} \sum_g \chi_S(g^2)$, where χ_S is the character of that sector, so $s = d_{\chi} \nu_{\chi}$: it equals $+d$ for real type, 0 for a merged conjugate pair and $-d$ for quaternionic type. Idempotency, integer trace, integer s and $\sum_S P_S = I$ are asserted numerically at construction. Consequently the logit decomposition $\ell = \sum_S P_S \ell$ is exact to floating point rather than an attribution heuristic — the property that motivates using this task family at all.

Probe protocol. The training split is divided into a fit set (80%) and a selection set (20%), with an independent partition per run. The ridge strength is chosen on the selection set and the probe is then refitted at that value on the whole training split; every other recipe is fixed in advance and reported separately rather than selected. Validation accuracy is logged through training so that learning curves can be drawn; it drives no selection decision. Split, initialisation and probe partition use three separate seed streams, so none is confounded with the others.

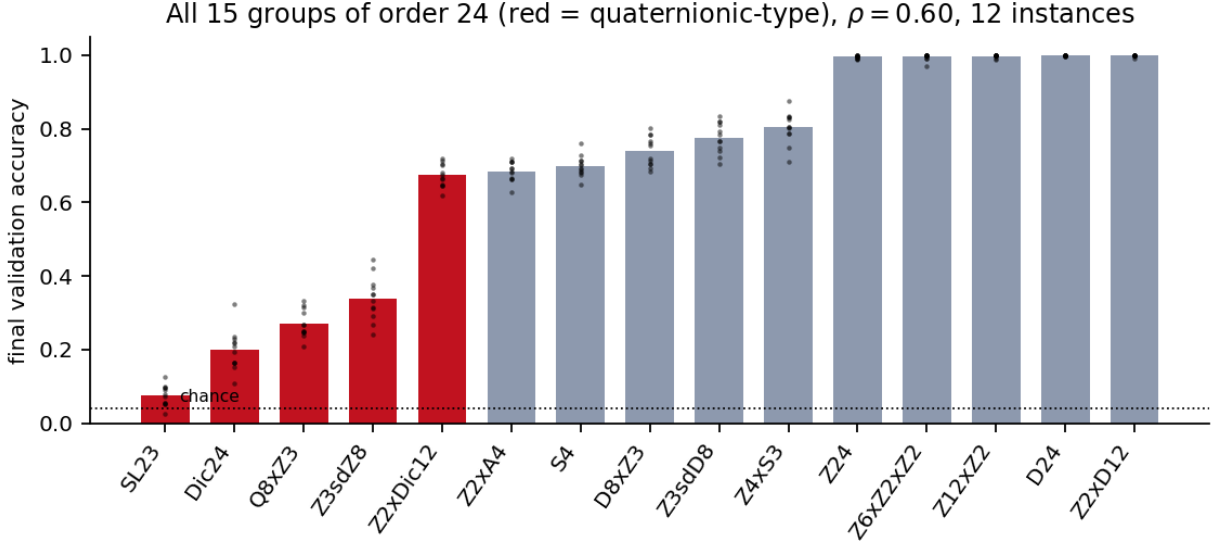


Figure 1: All fifteen groups of order 24 at $\rho = 0.60$, ranked by mean final validation accuracy. Red bars carry quaternionic-type irreducible representations. Dots are the twelve instances, each an independent relabelling, split and initialisation. Four of the five red bars are the bottom four; the five red bars are the bottom five with no inversion; the boundary is close, with $\mathbb{Z}_2 \times \text{Dic}_{12}$ at 0.674 sitting 0.011 below $\mathbb{Z}_2 \times A_4$ at 0.685.

Batching. Many runs are trained in one batched job. The cross-entropy is multiplied by the number of runs so that each run’s gradient equals what it would be if that run were trained alone. Without this the loss averages over $R \cdot N$ examples, every gradient is scaled by $1/R$, and Adam’s ϵ becomes an effective $R \cdot 10^{-8}$, comparable to $\sqrt{v}$ once gradients are small — exactly the phase these experiments measure.

3 Finding A: representation type predicts fixed-budget accuracy

3.1 The complete population

Figure 1 and Table 1 give final validation accuracy at $\rho = 0.60$ after 9k epochs, twelve instances per group, all fifteen groups of order 24.

3.2 Statistics, at the correct unit

The unit of replication is the group. Instances resample the same fifteen tasks; quoting instance counts would be pseudoreplication.

The preregistered primary test is the exact ordering statistic: that the five quaternionic-type groups occupy the bottom five positions. They do, with no inversion. Their rank sum is 15, the minimum attainable, and the worst non-quaternionic group ($\mathbb{Z}_2 \times A_4$ at 0.685) sits above the best quaternionic group ($\mathbb{Z}_2 \times \text{Dic}_{12}$ at 0.674). Under exchangeability of the fifteen groups,

$$\Pr(\text{the five occupy the bottom five}) = \binom{15}{5}^{-1} = \frac{1}{3003} = 3.3 \times 10^{-4}. \quad (2)$$

rank	group	q-mass	mean	sd	min	
1	$SL(2, 3)$	0.1667	0.075	0.028	0.026	quaternionic
2	Dic_{24}	0.5000	0.198	0.055	0.108	quaternionic
3	$Q_8 \times \mathbb{Z}_3$	0.1667	0.271	0.038	0.208	quaternionic
4	$\mathbb{Z}_3 \rtimes \mathbb{Z}_8$	0.1667	0.339	0.059	0.242	quaternionic
5	$\mathbb{Z}_2 \times Dic_{12}$	0.3333	0.674	0.031	0.619	quaternionic
6	$\mathbb{Z}_2 \times A_4$	0	0.685	0.027	0.628	
7	S_4	0	0.699	0.029	0.649	
8	$D_8 \times \mathbb{Z}_3$	0	0.739	0.040	0.684	
9	$\mathbb{Z}_3 \rtimes D_8$	0	0.776	0.041	0.706	
10	$\mathbb{Z}_4 \times S_3$	0	0.804	0.043	0.710	
11	$\mathbb{Z}_{24}$	0	0.995	0.004	0.987	
12	$\mathbb{Z}_6 \times \mathbb{Z}_2^2$	0	0.996	0.009	0.970	
13	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	0	0.998	0.004	0.987	
14	D_{24}	0	0.999	0.002	0.996	
15	$\mathbb{Z}_2 \times D_{12}$	0	0.999	0.003	0.991	

Table 1: All fifteen groups of order 24, $\rho = 0.60$, 9k epochs, twelve instances each. q-mass is the fraction of the regular representation carried by quaternionic-type sectors. Chance is $1/24 = 0.042$.

pair	non-quaternionic	quaternionic	Δ	Welch t
$D_8 \times \mathbb{Z}_3 / Q_8 \times \mathbb{Z}_3$	0.739	0.271	+0.469	29.3
D_{24} / Dic_{24}	0.999	0.198	+0.801	50.7
$\mathbb{Z}_2 \times D_{12} / \mathbb{Z}_2 \times Dic_{12}$ (<i>not a twin</i>)	0.999	0.674	+0.325	35.7

Table 2: The two character-table twin pairs at order 24 (top), and one degree- and class-size-matched contrast that is *not* a twin (bottom, italicised). Within a twin pair the ordinary complex character table is identical; $\mathbb{Z}_2 \times D_{12}$ and $\mathbb{Z}_2 \times Dic_{12}$ differ in their linear characters. $\rho = 0.60$.

The preregistered secondary analysis, an exact permutation test on group means over all 3003 labellings, gives $p = 3.3 \times 10^{-4}$ with a difference in means of +0.558.

Two caveats. Eight of the fifteen groups are the ones on which the hypothesis was formed, so this is a preregistered extension to the complete population rather than a wholly prospective confirmation; the genuinely unseen content is the seven new groups, of which the two quaternionic ones ($\mathbb{Z}_2 \times Dic_{12}$ and $\mathbb{Z}_3 \rtimes \mathbb{Z}_8$) both landed in the bottom five as predicted. And the margin at the boundary is 0.011, about a third of the within-group standard deviation there, so the ordering is exact but not comfortable.

Between-group variance is 0.101 and within-group variance 0.0011, a ratio of 92, where an instance varies relabelling, split and initialisation independently. Under this protocol, variation from those three sources is much smaller than variation between groups. This is a descriptive comparison of two sample variances, not a fitted variance-components decomposition.

3.3 Twin pairs

Table 2 and Figure 2 are the controlled contrast and the rank-sum test is not: comparing $\mathbb{Z}_{24}$ to S_4 compares two tasks that differ in everything.

There are exactly two character-table twin pairs at order 24, and both are among the groups on which the hypothesis was formed. They are therefore descriptive evidence, not a prospective test, and we report no sign test over them. The third row of Table 2 is a weaker contrast, matched on the

The two character-table twin pairs at order 24, and one degree- and class-size-matched contrast that is not a twin

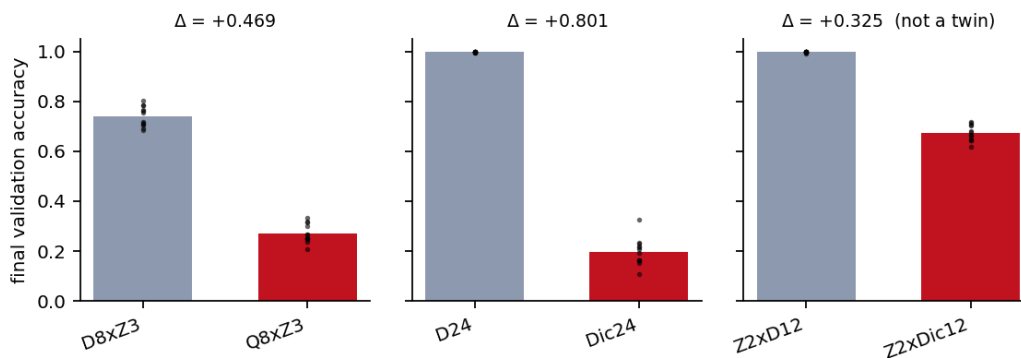


Figure 2: The two verified character-table twin pairs, and one degree- and class-size-matched contrast that is not a twin. Within a twin pair the ordinary complex character table is identical; the Frobenius–Schur indicator differs, along with the power map and everything else it determines.

multiset of irreducible degrees and the multiset of conjugacy-class sizes but *not* on the character table: $\mathbb{Z}_2 \times Dic_{12}$ has abelianisation $\mathbb{Z}_2 \times \mathbb{Z}_4$ and hence four linear characters taking the values $\pm i$, while every linear character of $\mathbb{Z}_2 \times D_{12}$ is real. It was preregistered and had never been trained, and it separates in the predicted direction, but it does not replicate the twin design. Improving on two twin pairs requires other orders.

3.4 q-mass is not a dose

Figure 3 shows the same effect as a dose–response on the original eight-group grid: the separation is in the training fraction required, not in learnability. The fraction of the regular representation carried by quaternionic sectors takes four values here. Mean accuracy by level is 0.869 ($q = 0$), 0.228 ($1/6$), 0.674 ($1/3$), 0.198 ($1/2$): correlation -0.697 , not monotone. This prediction was preregistered as risky and it failed. The presence of a quaternionic sector predicts the sign of the effect; q-mass does not scale it.

4 Finding B: what a linear-probe gap measures

Chou et al. [6] decompose a network into encoder and linear readout and define a sub-optimal-readout signature: linear-probe accuracy substantially exceeding model accuracy, indicating that the readout under-performs relative to what the encoder affords. They demonstrate it on permutation composition in S_5 on an MLP. The existence of such a gap is therefore not new.

Figure 4 shows the dynamics: in both twins the probe reaches ceiling long before the native readout, and in $Q_8 \times \mathbb{Z}_3$ the native readout never closes the distance. That a probe’s verdict depends on its capacity and regularisation is not itself new; it is the central point of the probing-control literature [8, 9]. What has not been examined is how much this particular grokking diagnostic depends on it. Table 3 reports nine non-redundant settings of the thirty readout configurations that `e3.probe_protocol.py` fits to *one unchanged* frozen $Q_8 \times \mathbb{Z}_3$ representation at $\rho = 0.70$: two Chou budgets, seven AdamW weight decays from a fresh initialisation, the same seven from the native readout, thirteen closed-form ridge strengths, and ridge with λ selected inside the training split. Every configuration reaches training accuracy 1.000 on the split it was fitted to: they all

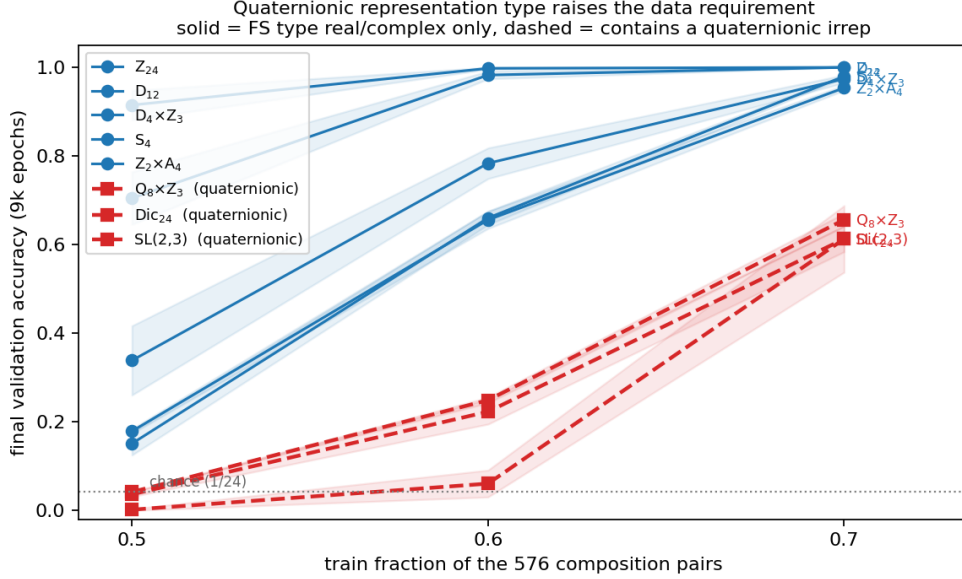


Figure 3: Dose-response on the original eight-group grid, two seeds, fixed 9k budget. Solid: no quaternionic sector. Dashed: quaternionic. The separation widens as ρ falls. In this released grid the quaternionic groups reach only 0.61–0.66 at $\rho = 0.70$ and no run crosses the 0.90 generalisation threshold, so these data show a higher data requirement but do not by themselves show that the tasks become easy at higher ρ . Produced by the original codebase from `results/dose_response.json`.

interpolate and differ only in which interpolating readout they select. Within a seed every recipe sees the same frozen representation; the three seeds are three separately trained representations and the table reports their mean.

The reported gap spans 0.256 (Figure 5). The endpoints differ in objective, optimiser, budget and initialisation as well as in regularisation, so this span measures sensitivity to the probe protocol as a whole; the clean regularisation-only contrast is the 0.179 span of §5, at a fixed representation and a fixed fitting procedure. Under the protocol of the paper that defines the signature — AdamW, $\text{lr } 10^{-3}$, weight decay 1.0, 200 full-batch epochs, justified there as ensuring that high probe accuracy reflects genuinely structured representations rather than memorisation — $Q_8 \times Z_3$ reports -0.021 , and at that weight decay run to convergence it reports $+0.002$: no readout defect. The character-table twins are also indistinguishable there ($+0.002$ against $+0.000$; three seeds, no equivalence test, so this is a failure to separate rather than a demonstration of equality). Under ridge with λ chosen inside the training split and refitted on all of it, the same frozen features give $+0.235$ against -0.033 and the twins separate. The decisive comparison is the weight decay itself: at $\lambda_{\text{probe}} = 1.0$ the gap is $+0.002$ and at $\lambda_{\text{probe}} = 0.01$ it is $+0.181$, with objective, optimiser, budget and initialisation all held fixed.

We do not claim that the weakly regularised readout is the “true” capability of the representation. A probe is always a measurement relative to a hypothesis class and an inductive bias, and [6] may reasonably intend to measure only especially accessible structure. The claim is narrower and, we think, unavoidable: *the linear-probe gap is not well defined without stating the probe’s regularisation*, and in this family the choice moves the reported value by more than a third of the accuracy range, including across zero and including the presence or absence of a twin-pair difference.

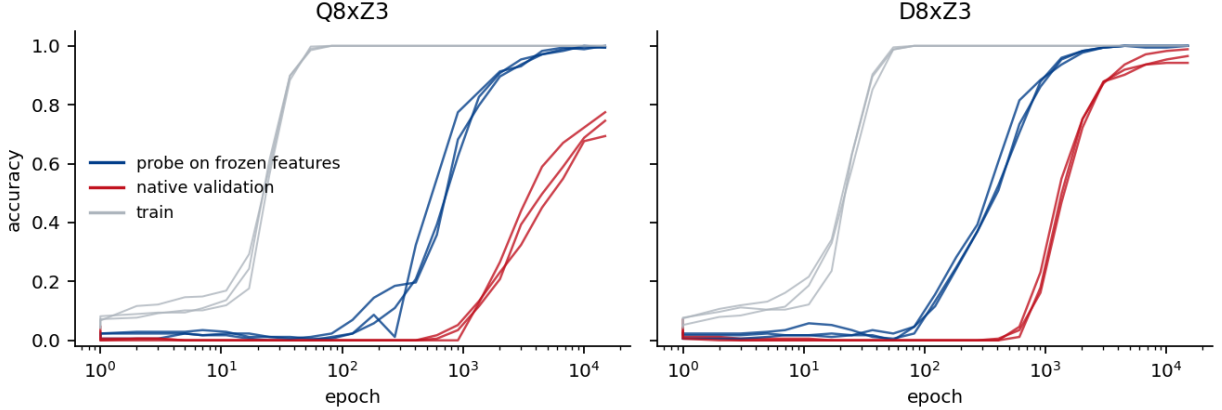


Figure 4: Two speeds, on the character-table twins. Grey is training accuracy, red the native readout on held-out data, blue a ridge probe fitted to the frozen hidden layer on the training split only with λ chosen inside that split. Three seeds each, $\rho = 0.70$, log epoch axis. In both groups the probe reaches ceiling long before the native readout does. In $D_8 \times \mathbb{Z}_3$ the native readout catches up; in $Q_8 \times \mathbb{Z}_3$ it does not, and the two curves stay separated for the rest of training. Generated by `analyse.py` from `results/e3_probe_protocol.json`.

Is this regime spurious? Chou et al. retire a published grokking example as spurious using a representation-degradation signature: probe accuracy falling while critical dimension rises, under a recipe that actively degrades representation quality. A referee will aim that at weight decay 1.0 at sub-critical ρ . We therefore run their discriminant on ourselves. Probe accuracy rises from near zero to 0.994–1.000 through training in every run and effective representation rank rises with it (57–60 to 69–71), so the degradation signature is absent in all six runs. The sub-optimal-readout signature is present in all three $Q_8 \times \mathbb{Z}_3$ seeds and in one of three $D_8 \times \mathbb{Z}_3$ seeds, so it is not cleanly absent in the grokking twin, only much rarer and much smaller.

5 Weight decay has two roles

Figure 6 plots the same data. An earlier version of this work titled this section “the obstruction is weight decay”. The factorial in Table 4 shows that phrase conflates two effects of opposite sign.

Read down the first column: as λ_{train} goes $0 \rightarrow 1$, the best achievable refit goes $0.013 \rightarrow 0.879$, a span of 0.865. Probe accuracy on the frozen layer goes $0.162 \rightarrow 0.992$ over the same range. Training weight decay is *associated with* greater linear decodability at this fixed budget; at $\lambda_{\text{train}} = 0$ there is nothing to express, and a probe–native difference measured there is a difference between two numbers near chance, not an unexpressed computation. For this reason we report native and probe accuracy as separate columns throughout and do not compare their difference across λ_{train} : the probe ceiling moves.

Read across the $\lambda_{\text{train}} = 1$ row: the refit goes $0.879 \rightarrow 0.734$, a span of 0.146, on a representation that does not change. Readout weight decay is what *limits expression*. At $\lambda_{\text{train}} = 1$ the native readout reaches 0.730; refitting at the training decay recovers +0.004, and refitting at zero recovers +0.148. The same contrast in the grokking twin $D_8 \times \mathbb{Z}_3$ is +0.002 and +0.023 from a native 0.975: there is little to recover when the readout is already expressing what the representation supports. Training accuracy is 1.000 in every cell: the representation is separable, many readouts interpolate the training split, and they generalise very differently.

readout recipe	$Q_8 \times \mathbb{Z}_3$ gap	$D_8 \times \mathbb{Z}_3$ gap
AdamW, wd 1.0, 200 epochs [6]	-0.021	-0.079
AdamW, wd 1.0, 4000 epochs	+0.008	+0.000
AdamW, wd 1.0, 4000 epochs, lr 2×10^{-3}	+0.002	+0.000
AdamW, wd 0.3	+0.033	+0.013
AdamW, wd 0.1	+0.125	+0.025
AdamW, wd 0.03	+0.170	+0.027
AdamW, wd 0.01	+0.181	+0.027
AdamW, wd 0	+0.177	+0.027
ridge, λ chosen inside train	+0.235	-0.033

Table 3: Probe accuracy minus native accuracy, one frozen representation, three seeds, $\rho = 0.70$. Training accuracy is 1.000 for every recipe. The top row is the protocol of [6], Appendix A.1.3. Every recipe is fitted on the model’s full training split, the same data the native readout saw.

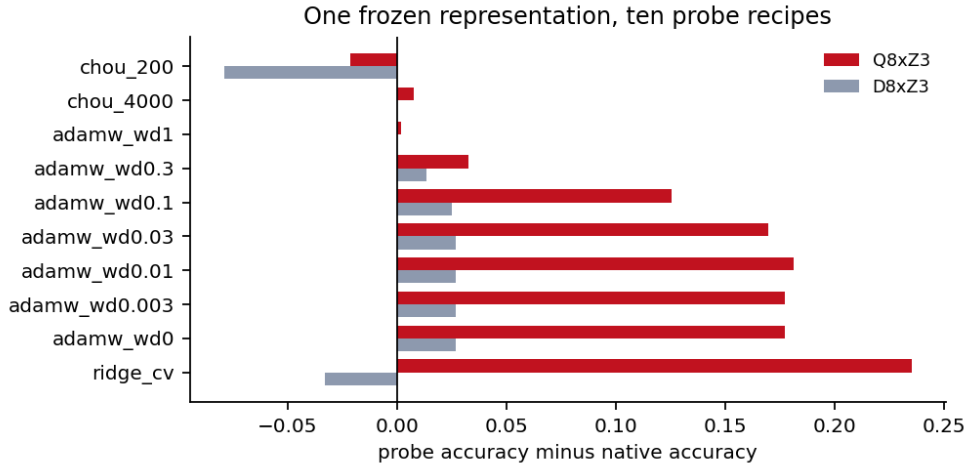


Figure 5: Probe accuracy minus native accuracy on one frozen representation that never changes, by readout recipe. Every recipe reaches training accuracy 1.000. `chou_200` is the protocol of [6] verbatim; it reports a negative gap, and the twins do not separate under it. `ridge_cv` reports +0.235 and separates them.

Across this grid, variation associated with training decay is substantially larger than variation associated with refit decay, and the two interact. We do not quote a ratio: the two spans are endpoint differences along one selected column and one selected row, so their quotient is descriptive geometry of this table rather than a factorial effect estimate. What matters is that both effects are real, they act in opposite directions, and a single unqualified statement about “weight decay” cannot describe them. This cuts against the standard account [1, 2, 5] — in which weight decay is the ingredient that causes grokking — only in the sense that the same term at the same strength does two things at two places in the network.

We have not established that the column effect is not a logit-scale effect. Truong [10] shows that under cross-entropy the weight-norm dependence of the grokking delay collapses onto the effective logit scale ($R^2 = 0.97$, with the norm dose adding 1–2%), and that shifting norm mass toward the readout at fixed total norm changes the delay by 1.7–2.2 $\times$. Our dependent variable is different — expression at a fixed budget rather than delay — and the direction is opposite, since higher weight

λ_{train}	λ_{readout}					
	0	0.01	0.1	0.3	1	3
0	0.013	0.012	0.012	0.002	0.002	0.000
0.01	0.023	0.015	0.013	0.006	0.000	0.000
0.1	0.276	0.266	0.150	0.143	0.123	0.044
0.3	0.798	0.796	0.736	0.586	0.539	0.561
1	0.879	0.881	0.842	0.761	0.734	0.765
3	0.821	0.815	0.769	0.667	0.640	0.672

Table 4: $Q_8 \times \mathbb{Z}_3$, $\rho = 0.70$, mean of three seeds — three independently trained representations, not one. Validation accuracy of a readout refitted on the frozen hidden layer, crossing the weight decay used end to end with the weight decay used in the refit.

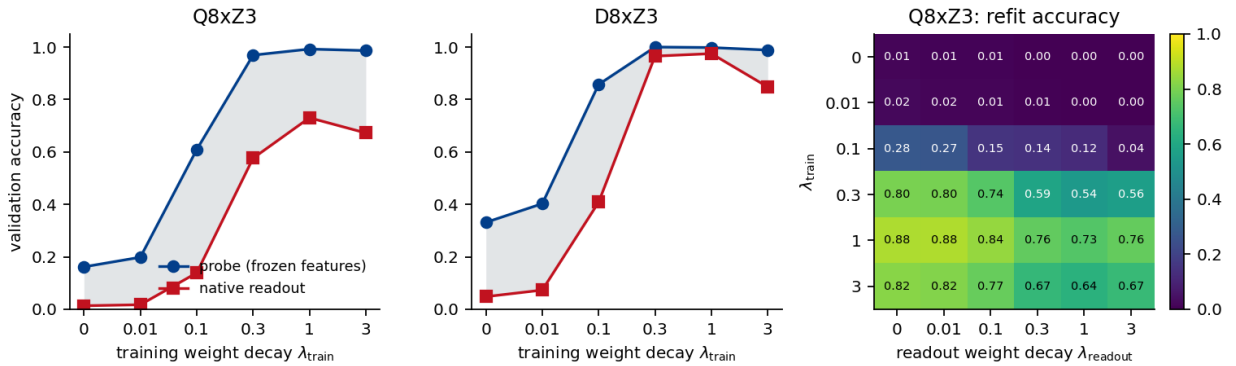


Figure 6: Left and centre: native and probe accuracy against end-to-end weight decay, reported separately rather than as a difference, because the probe ceiling moves. Right: the refit accuracy of Table 4 as a heatmap. Variation down the rows (training decay) is much larger than variation across the columns (refit decay), and the two act in opposite directions.

decay lowers the norm and hence the logit scale. Whether the column effect collapses onto the realised logit scale is a preregistered open question; the logit scale is logged at every checkpoint and the regression is outstanding.

6 Sector localisation is partial

Writing $W_2 = \sum_S P_S W_2$ exactly, we transplant a single sector’s readout block between the native and refitted output maps, holding W_1 frozen so there is no basis-alignment confound. The donor is fixed in advance ($\lambda_{\text{readout}} = 0$, native initialisation). An earlier version selected the donor by validation accuracy over eight configurations, which is selection on the reported metric; that result does not survive the fixed donor.

In Table 5 the two algebraically meaningful sectors carry comparable amounts and neither dominates; the difference between them is within three-seed noise. In the grokking twin the whole repair range is 0.031, too small for the fractions to carry meaning, so no localisation is claimed there. The group-equivariant control carries almost nothing, so the effect is not generic to any rank-8 equivariant subspace. This is weaker and less specific than an earlier version of this work claimed. It is localisation, not explanation.

sector	native + repaired	sufficiency	repaired + native	necessity
trace-8 complex pair	0.809	50.0%	0.827	37.8%
trace-4 Frobenius–Schur-typed	0.825	60.8%	0.817	44.6%
group-equivariant rank 8	0.738	0.0%	0.875	4.1%
random rank 8	0.119	—	0.089	—

Table 5: $Q_8 \times \mathbb{Z}_3$, mean of three independently trained representations, native 0.738, repaired 0.881. Sufficiency and necessity are fractions of the 0.143 repair range. A random rank-8 subspace of $\mathbb{R}^{24}$ is not group-equivariant and cuts across every isotypic component, so it destroys any working readout; it is shown for scale, not as a control.

One quantity is uninformative and we exclude it. The per-sector relative change $\|P_S(W_2^{\text{rep}} - W_2^{\text{nat}})\|/\|P_S W_2^{\text{nat}}\|$ reads 18–32 $\times$ for one one-dimensional sector in every run. This is division by noise: cross-entropy is invariant to a constant shift of all logits, so that direction is unconstrained and the denominator is near zero.

7 The trace-8 account does not explain Finding A

Neither D_{24} nor Dic_{24} possesses a trace-8 sector; all their isotypic sectors are trace 1 or trace 4. The readout defect of §4 therefore cannot operate through that sector in that pair, yet the accuracy separation appears there anyway, with three times the Frobenius–Schur contrast. This rules out one specific route between the two findings. It does not establish that they are independent: Finding B has not been measured in the D_{24} pair at all, and the trace-4 sector remains a plausible common mechanism.

A further obstacle to any unifying account: the Frobenius–Schur distinction lives in the trace-4 sector, and for $D_8 \times \mathbb{Z}_3$ and $Q_8 \times \mathbb{Z}_3$ the trace-8 sectors are isomorphic as real algebras (both $M_2(\mathbb{C})$, being the 2-dimensional irreducible representation tensored with the nontrivial cube roots of unity). Under the fixed donor the trace-4 sector now carries at least as much of the repair as the trace-8 sector, which weakens this obstacle but does not remove it, since the two are within noise of each other. Any mechanism proposed for (B) must address it.

8 Threats to validity

Character-table twins do not isolate Frobenius–Schur type. They match the ordinary complex character table. They do not match the power map, the involution count, the subgroup lattice, the automorphism structure or the element-order profile, all of which a network can read off the Cayley table. Involution count is the live competitor: D_{24} has 13 and Dic_{24} has 1, so it is anti-correlated with quaternionic type on these pairs by construction. The correct claim is that Frobenius–Schur type is a candidate predictor in the tested population, not an isolated cause. Separating the candidates requires an out-of-sample predictive comparison against the competing invariants, which is outstanding.

Sample size. Fifteen groups is fifteen groups, and one order. Two twin pairs is two, and there are no more at order 24. Twelve instances per group shrink within-group variance but do not increase n at the group level.

Preregistered predictions that failed. Monotonicity in q-mass failed, and the identity of the interfering sector was not predicted in advance. These are recorded with their outcomes in the repository’s preregistration file. Two further predictions — the alternative-predictor comparison and the logit-scale regression — have not yet been run.

Post-hoc sector selection. The interfering sector was identified by scanning all sectors and reporting the one that repaired. That selection is post-hoc. $SL(2, 3)$ replicates a repair with a *different* sector, which is honest evidence that which sector interferes is not yet predictable a priori. Until it is, no preregistered prediction of the sector identity is possible, and this remains the largest outstanding weakness in (B).

Single architecture. One-hidden-layer MLP, one optimiser, one recipe. A TransformerLens port trains but had not converged on the harder groups at the budgets tested; no architecture-transfer claim is made.

Validation set size. 173 examples at $\rho = 0.70$ and 231 at $\rho = 0.60$. Precision is limited accordingly.

Budget censoring. Non-generalising entries are lower bounds at the stated budget. Censoring at a single training fraction conflates “does not generalise” with “is below the critical dataset size”; the dense ρ sweep that would resolve this is outstanding.

9 Planned work

Each with a preregistered gate — the outcome that would falsify it — because the value of this task family is that such gates are cheap.

1. **Does q-mass beat the alternatives out of sample?** Leave-one-group-out prediction of mean accuracy from q-mass against involution count, exponent, centre order, abelianisation, class count, largest irreducible dimension, commuting probability and element-order profile. *Gate:* if involution count matches it, the claim narrows to “a power-map statistic predicts sample complexity”, with Frobenius–Schur type one such statistic among several.
2. **Does the expression gap collapse onto the logit scale?** Regress the gap on the realised logit scale, then add λ_{readout} . *Gate:* if the increment in R^2 is below 0.10, the readout defect belongs to [10] rather than here.
3. **Other orders.** Order 16 has character-table twins, as does order 48. *Gate:* if the twin separation does not replicate at another order, the effect is order-24-specific.
4. **Predict the interfering sector before running.** Formulate a rule from the $Q_8 \times \mathbb{Z}_3$, $D_8 \times \mathbb{Z}_3$ and $SL(2, 3)$ cases and preregister it on a held-out group. *Gate:* a wrong prediction retires the sector-level account and leaves the readout-level one, which is fine.
5. **Architecture transfer.** Train a transformer to convergence on the same task and measure the probe families. *Gate:* no transfer means the finding is MLP-specific. Lower priority than 1–4, because a mechanism should be understood before it is chased across architectures.

Deliberately not attempted. Constructing semisimple algebras $\mathbb{R}^4 \oplus \mathbb{C}^4 \oplus M_2(\mathbb{R}) \oplus M_2(\mathbb{C})$ versus $\mathbb{R}^4 \oplus \mathbb{C}^4 \oplus \mathbb{H} \oplus M_2(\mathbb{C})$ to isolate the Brauer class directly would be the cleanest possible test of (A). We do not attempt it here: basis elements of a general semisimple algebra are not closed under multiplication, so the task becomes structure-tensor regression rather than classification, changing the loss, the output format and plausibly the learning dynamics. It is a separate study, not an extension of this one.

10 Scope

This is a controlled algebraic testbed, not a claim about frontier models. Interpretability of small models on algorithmic tasks is a mature area in which the bar for a novel contribution is high, and nothing here should be assumed to transfer. What the setting buys is exactness: the projectors come from the Cayley table, sum to the identity, and make the decomposition verifiable to floating point rather than arguable. Settings in which a mechanistic claim can be verified rather than defended are rare.

The transferable question, if there is one, concerns measurement. A model can contain a computation that its own output map does not express. Whether a probe reports that depends on how the probe is regularised — by 0.256 of accuracy here across recipes, and 0.179 varying the regulariser alone, with a published protocol landing on the side that reports no defect and losing the twin-pair distinction with it. Distinguishing capability absent from capability present but unexpressed matters for evaluation and elicitation, and it cannot be done with an unspecified probe.

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